



SETP4583

SURFACTANT AND DETERGENT TECHNOLOGY

SECTION EB01

GROUP ASSIGNMENT:

INDIVIDUAL REFLECTIVE JOURNAL ASSIGNMENT

LECTURER:

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1.0 Introduction

Before enrolling in the Surfactant and Detergent Technology course, my understanding of hygiene products was purely that of a consumer. I used shampoo, body wash, and laundry detergent daily without giving a second thought to the complex chemistry inside the bottle. My initial expectation was that this course would be heavily theoretical, focusing solely on chemical structures and reaction mechanisms on paper. However, I was driven to take this course by a genuine curiosity about the products that play such a pivotal role in our daily lives. I wanted to learn the ingredient lists on the back of the bottles and understand exactly how a clear liquid turns into a cleansing foam.

My primary goal was to bridge the gap between my general interest in chemistry and practical application. I wanted to learn about the different types of surfactants such as anionic, cationic, non-ionic, and amphoteric, and understand their specific functions. More importantly, I hoped to gain the ability to look at a product not just for its brand name or fragrance, but for its functional formulation. I entered the course expecting to learn "what" ingredients are. I did not expect to learn "how" to actually be a creator and formulator myself.

2.0 Reflection on Key Learnings

The most interesting part of this course for me was definitely the practical side. Learning the theory in class was useful, but actually holding the chemicals and mixing them made a huge difference in my understanding.

In the lab, we got to handle raw materials like **SLES (Sodium Laureth Sulfate)**, **LABS (Linear Alkylbenzene Sulfonate)**, **Cocamidopropyl Betaine**, and **Berol**. It was a really special experience for me because, surprisingly, I hadn't stepped into a science lab to do experiments since I was in secondary school. Getting back into that environment and mixing things with my own hands made me feel like I was really doing science, not just studying it.

My biggest "aha" moment came when we were trying to make our group's body wash, "PURELUX." We took the chemicals like Betaine and SLES and tried to turn them into a soap. I realized that you can't just mix them randomly. For example, we learned that SLES is good for cleaning, but if you use it alone, it's too harsh. Adding Betaine is necessary to make it gentle. Watching the mixture change from watery chemicals into a thick, pearlescent soap

was really satisfying. I also learned about the "cold mix" process, where we didn't use heat. It was cool to see how we could make a product just by stirring at room temperature if we used the right ingredients.



Figure 2.1&2.2: Stirring the Chemical Surfactants in Lab

3.0 Application and Real-World Relevance

The knowledge gained from this course has fundamentally changed how I interact with the real world, specifically in how I shop for and view consumer goods. The most immediate application of this knowledge is my new habit of reading ingredient lists. Previously, looking at the back of a shampoo bottle was like reading a foreign language. Now, I can scan a label and immediately identify the functional components. When I see "Sodium Laureth Sulfate" listed first, I know it is the primary cleansing agent. When I see "Cocamidopropyl Betaine," I understand it is there to mitigate irritation.

This skill extends beyond personal curiosity, it allows me to critically evaluate marketing claims. For example, when a product claims to be "gentle" or "baby-safe," I now look for the presence of mild amphoteric surfactants rather than trusting the slogan blindly. I can also assess the "value" of a product. I realized that many expensive brands use the exact same base surfactant system as cheaper alternatives, differing only in fragrance or marketing budget.

In a broader industry context, this course has highlighted the balance companies must strike between cost, performance, and sustainability. During our project, we chose a "cold mix" process to save energy and focused on biodegradable ingredients. This mirrors the current shift in the global detergent industry towards "green chemistry." I now understand that "sustainability" in detergents isn't just about the packaging; it's about choosing surfactants like Betaine that degrade easily in wastewater treatment, versus older, harsher chemicals that might linger in the environment. This connection between the choices we made in the lab and the global environmental standards has shown me the immense responsibility formulators hold in protecting our ecosystems.

4.0 Challenges and Growth

Throughout the semester, I encountered several challenges that facilitated personal and academic growth. A significant technical challenge arose during the product development phase. Producing a stable detergent required precise methodology; initially, our group struggled with achieving the correct viscosity and texture. This difficulty underscored the importance of accuracy in scientific experimentation and required us to refine our formulation process through trial and error.

Additionally, the soft skills component of the course presented its own set of challenges, particularly regarding the group project and final presentation. Collaborating effectively with team members to align on a single product vision required negotiation and compromise. We had to ensure that every member contributed to the "PURELUX" concept, from ingredient selection to packaging design. Furthermore, presenting technical data confidently was an area I initially found daunting. However, by thoroughly researching the scientific rationale behind our formulation, I was able to overcome this anxiety. This experience taught me that thorough preparation and a deep understanding of the subject matter are essential for effective technical communication.

5.0 Personal and Professional Development

Reflecting on the past 14 weeks, this course has significantly influenced my professional mindset. It has shifted my perspective from being a passive learner to an active problem solver. I used to think of chemistry as just another subject to study. However, I now recognize it as a way to create practical solutions, whether it is removing stains or developing skin-safe products.

From a professional perspective, this course opened my eyes to how products are created from scratch. Although I might not work in a lab forever, understanding the whole process is invaluable. I now know what it takes to launch a product, including selecting raw materials and passing safety tests. This is a crucial skill set that I can use in any future job related to product development.

In the future, I want to learn more about the business side of this industry. I am especially curious about how factories take a small lab experiment and produce it on a large scale without lowering the quality. Overall, this course gave me the confidence to work in a lab, read technical documents, and work well in a team. I believe these skills will be very helpful in my future career.



Figure 5.1 Group Photo of the Last Class